Slikovna Biometrija – Exam 2017

1. Name and explain briefly each of the authentication types.

- *Pair Matching* → Do these to images match?
- *Verification* → Does this sample of Mr.Bean match with the one in our system? (1:1)
- *Identification* → Does this sample exist in our system? (1:n)
- *Negative Authentication* → I'm not the subject X/ I'm not part of group X
- *Deduplication* → Do we have to enroll you?

2. At which stage in the recognition pipeline do we use the similarity score and the distance score? What is the difference between the similarity score and the distance score?

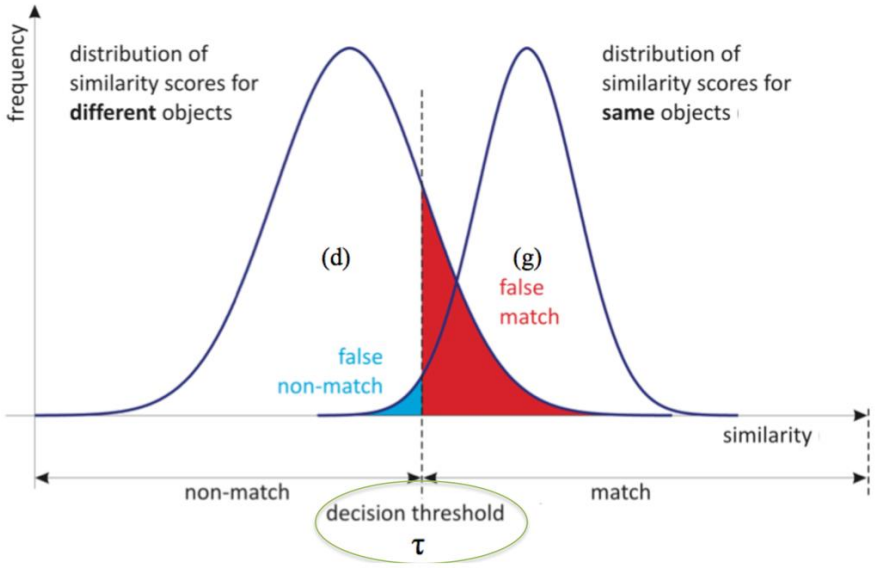
The similarity score and the distance score are used in the matching stage of the recognition pipeline. We want the similarity score to be as high as possible but we want the distance score to be as low as possible. However if the distance score is equal to 0 we should rather view it as an impostor because of the really low probability of that actually happening.

3. Explain how do we calculate False Non-Match Rate? Also, sketch and mark it using the genuine and impostor distributions.

If we have know the function g we can use:

$$g_{FNM}(\tau) = \int_{-\infty}^{\tau} g(s)ds$$

But because we probably don't, we calculate it by using the following formula:
FNMR = num of false non-matches for threshold T / num of all genuine comparisons



4. Explain measures AUC and EER (ERR?). What does it mean if AUC is 0.5?

AUC → it is also possible to compare the performance of the two systems by estimating the area under the ROC curve (AUC). The value of AUC ranges from 0.5 to 1, with 1 indicating that there are no errors.

EER → The performance of a biometric system may also be summarized using other single-valued measures such as the Equal Error Rate (EER). The EER refers to that point in a DET (or ROC) curve where the FAR equals the FRR; a lower EER value, therefore, indicates better performance.

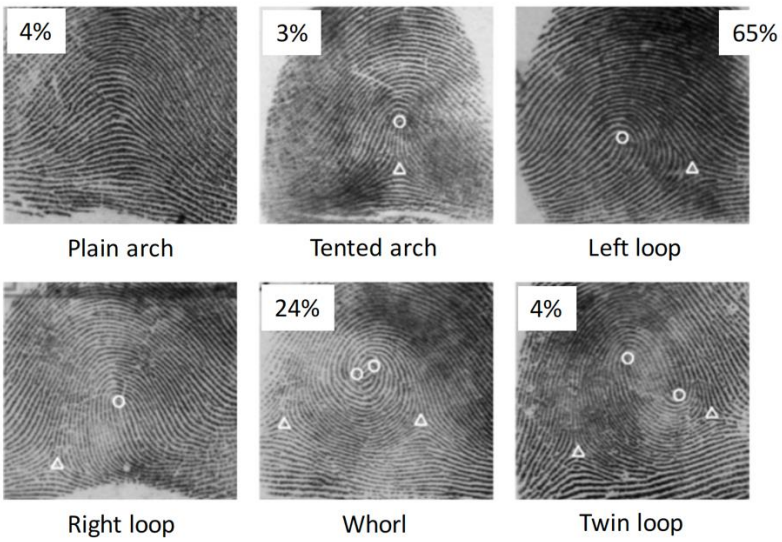
AUC = 0.5 → worst case scenario, our biometric system is trash.

5. Which factors must be considered to determine the suitability of a physical or a behavioral trait?

- Universality – possess the trait – minimize failure to enroll
- Uniqueness – sufficiently different traits – the FMR should always be sufficiently low
- Permanence – invariant over a period of time – if not it may lead to high FNMR
- Measurability – possible to acquire and digitize – can we measure something on top of our modality? The data we acquire should let us do that.
- Performance – metric + resources & throughput – how many samples can we process in a period of time
- Acceptability – willingness to present the trait – Indians don't accept shaking hands
- Circumvention (izogibanje) – imitate traits / evade recognition

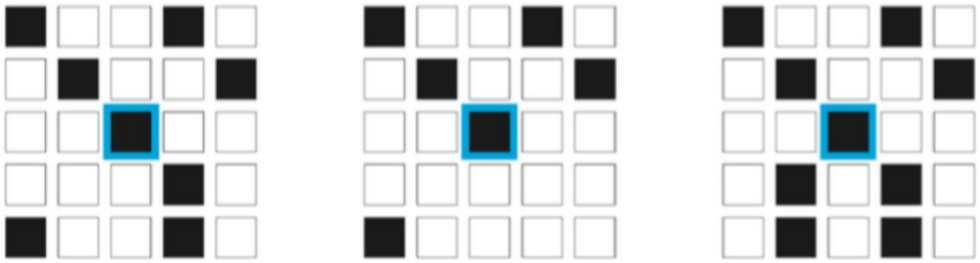
6. Name the fingerprint pattern types that are based on spatial relationship between singular points. To which type the majority of prints belong to? What is the expected number of singular points within a fingerprint?

Even more abstract representation is the pattern type (fingerprint class) → Deduced based on the number of loops and deltas and the spatial relationship between them.



7. How do we extract minutiae from binarised and thinned fingerprint image?

Ridge pixels with 3 ridge pixel neighbours are identified as ridge **bifurcations.**
Those with only one ridge pixel neighbour are identified as ridge **endings.**



#ridge_neighbors=2 => ridge (no minutiae)

=1 => ridge ending

=3 => bifurcation

8. How many pixels across the iris in the radial direction for successful processing do most iris recognition systems expect? Why do we need to use near-IR spectrum for iris acquisition?

Most iris recognition systems expect 100 - 200 pixels across the iris in the radial direction for successful processing.
Using near-IR spectrum allows usable texture imaging even for dark irises. This also prevents the percentage of FMR to rise.

9. In face detection step we used integral image. Why do we need it and how do we calculate it? How many cascades do we need to reach a GMR of 0.9 and a FMR of approximately 10−6, if each cascade has a GMR of 0.99 and a FMR of 0.3?

Set of Haar-like filters of different sizes need to be applied to each window and the filter responses must be combined in an appropriate way in order to detect a face → more than 180,000 different Haar-like features can be derived from a detection window of size 24x24 pixels → **to reduce computational burden calculate integral image**

The integral represents the sum of all pixel intensities above and to the left of (x,y) in the original image.

Input image

1	2	2
2	3	0
1	0	3

→

1	3	5
3	8	10
4	9	14

Integral image

We need 10 stages: (0.9910 ≈ 0.9) and (0.310 ≈ 6x10-6).

10. What is the difference between Eigenfaces and Fisherfaces? Which methods are behind each of them?

Eigenfaces attempt to maximize the scatter of the training images in faces space (PCA method)
Fisherfaces attempt to maximize the between class scatter, while minimizing the within class scatter (LDA method)

Slikovna Biometrija – Exam 2018

1. Which of the statements are scientific facts?

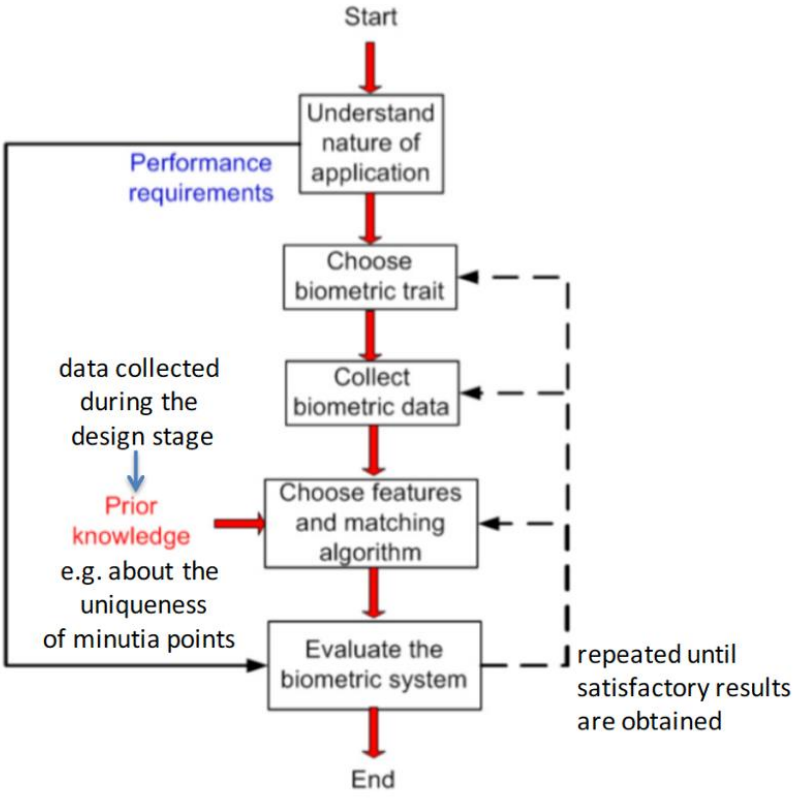
a. **The physical trait itself is unique – NOT A FACT**
b. **The physical trait itself is permanent – NOT A FACT**
c. **Biometric systems rely only on the digital measurments and not on the real physical traits – TRUE FACT**
d. **Sensing introduces variations in the samples of the same biometric trait of a user obtained over a period of time – TRUE FACT**
e. **If two feature sets are identical, it may be a strong indication that the biometric data actually comes from an adversary (nasprotnik) who is replaying the data at an earlier time – TRUE FACT**

2. Name problems with pertaining to higher error rates (FNMR)?

3. Explain the ROC curve and what it reveals if we compute two systems?

The FAR and FRR at different values of threshold are measured and summarized in the form of curves.
The best way to compare the performance of two biometric systems is to examine their ROC curves.
If the FRR of one biometric system (A) is
consistently lower than the FRR of the other system (B) for corresponding values of FAR, one can conclude that the matching performance of
biometric system A is better than that of B

4. Name and sketch general steps in the design cycle of a biometric system.



5. Which are the levels of features in fingerprints; name them by features? Define the meaning of singular points. What kind of singular points types are we looking for within the image?

L1 level features → orientation map and singular points
L2 level features → ridge skeleton image – consists of minutiae points
L3 level featues → extra detailed – inner holes (sweat pores) and outer contours (edges)

We are looking for two basic types – a loop and a delta type

6. Name fingerprint scan sensors technology. Describe one of them in detail.

- Optical FTIR-based sensor
- Capacitance-based sensor
- Ultrasound reflection-based sensor
- Piezoelectric effect-based sensor
- Temperature differential-based sensor

Capacitance-based sensor → it is more commonly used and very small in size. Represented with an array of electordes. Fingerprint skin acts as the other electrode, forming a miniature capacitor. The magnitude of electrical charges depends on the distance between the fingerprint surface and the capacitance plates. Capacitance due to the ridges is higher than those formed by valleys. Proper grounding needed due to tip electrostatic discharges

7. **Sketch basic taxonomy of fingerprint quality evaluation algorithms. Explain briefly how NFIQ algorithm works.**

Prvi del še moreš – kaj je taxonomy.

NFIQ:

- Combines local and global properites
- Prediction of matching efficiency
- For each feature (minutiae) calculate ridge density and clarity, high curvature areas, dominant block orientation, etc.
- Evaluate each block based on features quality (4 – perfect q., ... , 0 - background)
- Get overall quality (class 1 – perfect q., ... , class 5 – bad q.)
- Registration with only class 1 and class 2 quality is acceptable

8. **How big is tipically the iris code? Why?**

The standard commercial iris code is 2048 bits (plus 2048-bit mask).
Why?

9. **When it comes to faces, what is the basic resemblance between how do people recognize faces and how the CNN recognizes them?**

10. **Describe the properties of multimodal systems.**

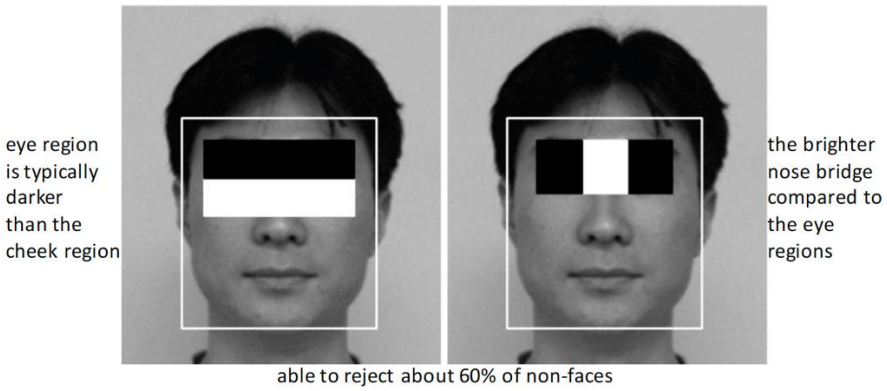
Slikovna Biometrija – Exam 2020

1. **Describe decision-level fusion. In this context describe majority voting method.**
2. **Desribe the properties of the multimodal biomettic systems.**
3. **What can you say about the analysis of the discriminative information contained in X layer of a CNN?**
4. **Whait is the main difference between PCA and LDA? How do Fisherfaces and Eigenfaces correlate to them? What should we do if the number of training samples (N) is smaller than the dimensionality of the data (d)?**
5. **Describe aligment by geometric warping. Any problem with this method? If yes, what kind of a problem and how can we mitigate it?**

Basic idea → rotate the face around one eye
Input → eyes positions & output parameters: horizontal / vertical offset, size
Output → scaled, rotated and cropped image
Problem → yes, only in-plane rotation addressed
Mitigate it → by using facial landmark detection. Find location of facial landmarks to align

6. How and why do we use cascades in the Viola-Jones method? Which are the two most effective Haar-like filters? In which stage of the cascades would you use them?

Classifiers with a simple & smaller set of filters an be first used to screen out a large number of non-face images in the early stage and then more powerful classifiers can be used in the later stage. **Boljša razlaga?**



able to reject about 60% of non-faces

7. Describe and link the following terms together: Singular points, pattern type, minutiae, resolution.

When talking about L1 fingerprint features we talk about singular points. Singular points are an abstract representation of the pattern type (fingerprint class) → the pattern type is deduced based on the number of loops and deltas and the spatial relationship between them.
However if our sensor is capable of capturing an image with a resolution 500 ppi or more, we can analyse and detect L2 features which are the minutiae points.

8. What kind of image imperfections can we find by analysing the Fourier spectra of local iris regions? Describe the properties of the Fourier spectra for each of them.

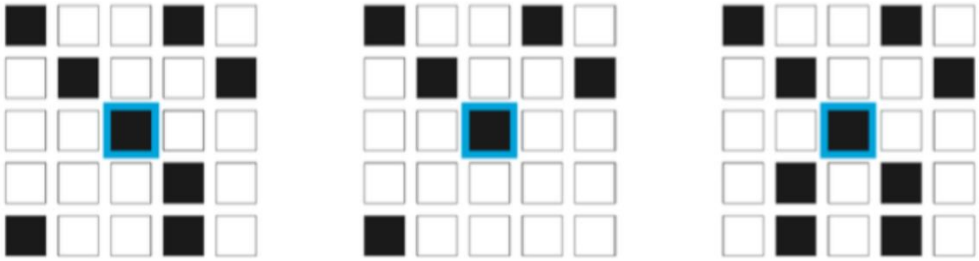
- **Out of focus blur** → Fourier spectrum is supposed to be largley dominated by the low frequency components.
- **Motion blur** → lacks middle and high frequency components and has frequency distribution similar to that of an out-of-focus image.
- **Occlusion due to the eyelashes, eyelids** → contains significant middle and high frequency components

9. Why do we need to normalise the iris image? Describe the normalisation process. Why do we need the iris mask for?

Addresses the problem of variations in pupil size across multiple images, subjects, sensors, imaging distances (main concern: dilation, contraction).
We use the Daugman's rubber sheet model → linearly stretching or compressing the imaged iris to a standard size frame.
We use a mask to mask out the "noise" in the image – that being the eyelids, eyelashes,… We don't what them to impact our feature extraction.

10. How do we extract minuitae from (one-pixel wide) ridges? How do we determine their directions?

Ridge pixels with 3 ridge pixel neighbours are identified as ridge **bifurcations.**
Those with only one ridge pixel neighbour are identified as ridge **endings.**



#ridge_neighbors=2 => ridge (no minutiae)

=1 => ridge ending

=3 => bifurcation

Minutiae direction:

- **Ending** → trace the associated ridge to a fixed distance (say 10 pixels)
- **Bifurcation** → obtain three points by tracing the ridges to a fixed distance & calculate mean of the two directions, which difference is the smallest among the three ridges